$3d^6$

August 25, 2026

1 $\left|3d_{z^2}^2 3d_{x^2-y^2}^2 3d_{xz}^\alpha 3d_{xy}^\alpha 3d_{yz}^0\right\rangle$

Here are the CSFs:

$$|1,2,2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|2,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|3,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|4,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|5,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$
$$\begin{aligned}|6,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\ &+ \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle\end{aligned}$$
$$\begin{aligned}|7,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\ &+ \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle\end{aligned}$$
$$\begin{aligned}|8,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\ &+ \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle\end{aligned}$$
$$\begin{aligned}|9,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle \\ &+ \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle\end{aligned}$$

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\begin{aligned}
|10,2,1\rangle =&\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|11,2,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|12,2,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|13,2,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|14,2,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|15,2,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|16,2,-1\rangle =&\frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
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\begin{aligned}
|17,2,-1\rangle =&\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|18,2,-1\rangle =&\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|19,2,-1\rangle =&\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|20,2,-1\rangle =&\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
&+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
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$$|21,2,-2\rangle=\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|22,2,-2\rangle=\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|23,2,-2\rangle=\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|24,2,-2\rangle=\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|25,2,-2\rangle=\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|26,1,1\rangle=+\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|27,1,1\rangle=+\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|28,1,1\rangle=+\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|29,1,1\rangle=+\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|30,1,1\rangle=+\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$
\begin{aligned}
|31,1,0\rangle =&+\frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&-\frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

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\begin{aligned}
|32,1,0\rangle =& +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& -\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|33,1,0\rangle =& +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& -\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|34,1,0\rangle =& +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& -\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|35,1,0\rangle =& +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
& -\frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|36,1,-1\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
|37,1,-1\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
|38,1,-1\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
|39,1,-1\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
|40,1,-1\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
|41,1,1\rangle =& +\sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& -\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|42,1,1\rangle =& +\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& -\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|43,1,1\rangle =& +\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& -\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
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$$\begin{aligned}
|44,1,1\rangle =& + \sqrt{\frac{2}{3}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|45,1,1\rangle =& + \sqrt{\frac{2}{3}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle \\
& - \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle \\
|46,1,0\rangle =& \frac{1}{\sqrt{3}} \left|3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}} \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|47,1,0\rangle =& \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|48,1,0\rangle =& \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|49,1,0\rangle =& \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|50,1,0\rangle =& \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle \\
& + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}$$

$$|51,1,-1\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle$$

$$|52,1,-1\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle$$

$$|53,1,-1\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle$$

$$|54,1,-1\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|55,1,-1\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|56,1,1\rangle = \frac{\sqrt{3}}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|57,1,1\rangle = \frac{\sqrt{3}}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|58,1,1\rangle = \frac{\sqrt{3}}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|59,1,1\rangle = \frac{\sqrt{3}}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$
\begin{aligned}
|60,1,1\rangle =& \frac{\sqrt{3}}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
& - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|61,1,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|62,1,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|63,1,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|64,1,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|65,1,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|66,1,-1\rangle =& \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
& + \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$|67,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|68,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|69,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|70,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|71,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|72,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|73,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|74,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|75,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|76,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|77,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|78,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|79,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|80,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|81,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|82,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|83,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|84,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|85,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|86,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|87,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|88,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|89,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|90,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|91,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|92,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|93,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|94,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|95,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|96,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|97,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|98,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|99,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|100,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|101,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|102,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|103,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|104,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|105,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|106,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|107,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|108,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|109,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|110,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|111,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|112,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|113,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|114,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|115,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|116,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|117,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|118,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|119,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|120,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|121,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|122,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|123,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|124,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|125,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|126,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|127,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|128,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|129,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|130,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|131,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|132,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|133,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|134,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|135,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|136,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|137,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|138,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|139,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|140,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|141,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|142,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|143,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|144,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|145,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|146,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|147,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|148,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|149,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|150,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|151,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|152,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|153,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|154,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|155,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|156,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|157,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|158,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|159,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|160,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|161,0,0\rangle = +\frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|162,0,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|163,0,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|164,0,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|165,0,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|166,0,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|167,0,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|168,0,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$
\begin{aligned}
|169,0,0\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|170,0,0\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$|171,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|172,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|173,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|174,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|175,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|176,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|177,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|178,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|179,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|180,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|181,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|182,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|183,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|184,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|185,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|186,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|187,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|188,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|189,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|190,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|191,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|192,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|193,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|194,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|195,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|196,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|197,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|198,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|199,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|200,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|201,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|202,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|203,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|204,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|205,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|206,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|207,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|208,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|209,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|210,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 94 124 154

$\langle 94,1,1\,|L_x|\,29,1,1\rangle = \frac{1}{\sqrt{2}}I$
$\langle 94,1,1\,|L_x|\,30,1,1\rangle = \sqrt{\frac{3}{2}}I$
$\langle 94,1,1\,|L_x|\,44,1,1\rangle = -\frac{1}{\sqrt{6}}I$
$\langle 94,1,1\,|L_x|\,45,1,1\rangle = -\frac{1}{\sqrt{2}}I$
$\langle 94,1,1\,|L_x|\,59,1,1\rangle = -\frac{2}{\sqrt{3}}I$
$\langle 94,1,1\,|L_x|\,60,1,1\rangle = -2I$

Δg_{xx}

$+\frac{1}{2}\zeta\Delta_{94-29}^{-1}$
$+\frac{3}{2}\zeta\Delta_{94-30}^{-1}$
$+\frac{1}{6}\zeta\Delta_{94-44}^{-1}$
$+\frac{1}{2}\zeta\Delta_{94-45}^{-1}$
$+\frac{4}{3}\zeta\Delta_{94-59}^{-1}$
$+4\zeta\Delta_{94-60}^{-1}$

$\langle 94,1,1\,|L_y|\,81,1,1\rangle = I$
$\langle 94,1,1\,|L_y|\,82,1,1\rangle = -\sqrt{3}I$
$\langle 94,1,1\,|L_y|\,91,1,1\rangle = I$

Δg_{yy}

$+\zeta\Delta^{-1}_{94-81}$
$+3\zeta\Delta^{-1}_{94-82}$
$+\zeta\Delta^{-1}_{94-91}$

$\langle 94,1,1\,|L_z|\,92,1,1\rangle = 2I$
$\langle 94,1,1\,|L_z|\,100,1,1\rangle = -I$

Δg_{zz}

$+4\zeta\Delta^{-1}_{94-92}$
$+\zeta\Delta^{-1}_{94-100}$

For a multiplicity of 3 we have the following:

$$D = \langle 1,1|1,1\rangle - \langle 1,0|1,0\rangle$$

$$E = \langle 1,-1|1,1\rangle$$

$$\begin{aligned}
\left\langle 1,1\left|\hat{H}_{eff}\right|1,1\right\rangle = &-\zeta^2\,(\\
&+\frac{1}{4}\Delta^{-1}_{4} + \frac{3}{4}\Delta^{-1}_{5} + \frac{1}{24}\Delta^{-1}_{14} + \frac{1}{8}\Delta^{-1}_{15} + \frac{1}{16}\Delta^{-1}_{34} + \frac{3}{16}\Delta^{-1}_{35} \\
&+\frac{1}{48}\Delta^{-1}_{49} + \frac{1}{16}\Delta^{-1}_{50} + \frac{1}{24}\Delta^{-1}_{64} + \frac{1}{8}\Delta^{-1}_{65} + \Delta^{-1}_{92} + \frac{1}{4}\Delta^{-1}_{100} \\
&+\frac{1}{8}\Delta^{-1}_{111} + \frac{3}{8}\Delta^{-1}_{112} + \frac{1}{8}\Delta^{-1}_{121} + \frac{1}{16}\Delta^{-1}_{164} + \frac{3}{16}\Delta^{-1}_{165} + \frac{1}{48}\Delta^{-1}_{169} \\
&+\frac{1}{16}\Delta^{-1}_{170} + \frac{1}{8}\Delta^{-1}_{181} + \frac{3}{8}\Delta^{-1}_{182} + \frac{1}{8}\Delta^{-1}_{191} + \frac{1}{4}\Delta^{-1}_{206} + \frac{1}{4}\Delta^{-1}_{210} \\
&)
\end{aligned}$$

$$\left\langle 1,0 \middle| \hat{H}_{eff} \middle| 1,0 \right\rangle = -\zeta^2 \Big(+ \frac{1}{8}\Delta_9^{-1} + \frac{3}{8}\Delta_{10}^{-1} + \frac{1}{8}\Delta_{19}^{-1} + \frac{3}{8}\Delta_{20}^{-1} + \frac{1}{16}\Delta_{29}^{-1} + \frac{3}{16}\Delta_{30}^{-1} + \frac{1}{16}\Delta_{39}^{-1} + \frac{3}{16}\Delta_{40}^{-1} + \frac{1}{48}\Delta_{44}^{-1} + \frac{1}{16}\Delta_{45}^{-1} + \frac{1}{48}\Delta_{54}^{-1} + \frac{1}{16}\Delta_{55}^{-1} + \frac{1}{24}\Delta_{59}^{-1} + \frac{1}{8}\Delta_{60}^{-1} + \frac{1}{24}\Delta_{69}^{-1} + \frac{1}{8}\Delta_{70}^{-1} + \frac{1}{8}\Delta_{81}^{-1} + \frac{3}{8}\Delta_{82}^{-1} + \frac{1}{8}\Delta_{91}^{-1} + \frac{1}{8}\Delta_{141}^{-1} + \frac{3}{8}\Delta_{142}^{-1} + \frac{1}{8}\Delta_{151}^{-1} + \Delta_{192}^{-1} + \frac{1}{4}\Delta_{200}^{-1} \Big)$$

$$\left\langle 1,-1 \middle| \hat{H}_{eff} \middle| 1,1 \right\rangle = -\zeta^2 \Big(- \frac{1}{24}\Delta_{14}^{-1} - \frac{1}{8}\Delta_{15}^{-1} + \frac{1}{16}\Delta_{34}^{-1} + \frac{3}{16}\Delta_{35}^{-1} + \frac{1}{48}\Delta_{49}^{-1} + \frac{1}{16}\Delta_{50}^{-1} + \frac{1}{24}\Delta_{64}^{-1} + \frac{1}{8}\Delta_{65}^{-1} - \frac{1}{8}\Delta_{111}^{-1} - \frac{3}{8}\Delta_{112}^{-1} - \frac{1}{8}\Delta_{121}^{-1} - \frac{1}{16}\Delta_{164}^{-1} - \frac{3}{16}\Delta_{165}^{-1} - \frac{1}{48}\Delta_{169}^{-1} - \frac{1}{16}\Delta_{170}^{-1} + \frac{1}{8}\Delta_{181}^{-1} + \frac{3}{8}\Delta_{182}^{-1} + \frac{1}{8}\Delta_{191}^{-1} - \frac{1}{4}\Delta_{206}^{-1} - \frac{1}{4}\Delta_{210}^{-1} \Big)$$

$$
\begin{aligned}
D = \zeta^2 \big(& \\
& - \frac{1}{4}\Delta_{4}^{-1} - \frac{3}{4}\Delta_{5}^{-1} + \frac{1}{8}\Delta_{9}^{-1} + \frac{3}{8}\Delta_{10}^{-1} - \frac{1}{24}\Delta_{14}^{-1} - \frac{1}{8}\Delta_{15}^{-1} \\
& + \frac{1}{8}\Delta_{19}^{-1} + \frac{3}{8}\Delta_{20}^{-1} + \frac{1}{16}\Delta_{29}^{-1} + \frac{3}{16}\Delta_{30}^{-1} - \frac{1}{16}\Delta_{34}^{-1} - \frac{3}{16}\Delta_{35}^{-1} \\
& + \frac{1}{16}\Delta_{39}^{-1} + \frac{3}{16}\Delta_{40}^{-1} + \frac{1}{48}\Delta_{44}^{-1} + \frac{1}{16}\Delta_{45}^{-1} - \frac{1}{48}\Delta_{49}^{-1} - \frac{1}{16}\Delta_{50}^{-1} \\
& + \frac{1}{48}\Delta_{54}^{-1} + \frac{1}{16}\Delta_{55}^{-1} + \frac{1}{24}\Delta_{59}^{-1} + \frac{1}{8}\Delta_{60}^{-1} - \frac{1}{24}\Delta_{64}^{-1} - \frac{1}{8}\Delta_{65}^{-1} \\
& + \frac{1}{24}\Delta_{69}^{-1} + \frac{1}{8}\Delta_{70}^{-1} + \frac{1}{8}\Delta_{81}^{-1} + \frac{3}{8}\Delta_{82}^{-1} + \frac{1}{8}\Delta_{91}^{-1} - \Delta_{92}^{-1} \\
& - \frac{1}{4}\Delta_{100}^{-1} - \frac{1}{8}\Delta_{111}^{-1} - \frac{3}{8}\Delta_{112}^{-1} - \frac{1}{8}\Delta_{121}^{-1} + \frac{1}{8}\Delta_{141}^{-1} + \frac{3}{8}\Delta_{142}^{-1} \\
& + \frac{1}{8}\Delta_{151}^{-1} - \frac{1}{16}\Delta_{164}^{-1} - \frac{3}{16}\Delta_{165}^{-1} - \frac{1}{48}\Delta_{169}^{-1} - \frac{1}{16}\Delta_{170}^{-1} - \frac{1}{8}\Delta_{181}^{-1} \\
& - \frac{3}{8}\Delta_{182}^{-1} - \frac{1}{8}\Delta_{191}^{-1} + \Delta_{192}^{-1} + \frac{1}{4}\Delta_{200}^{-1} - \frac{1}{4}\Delta_{206}^{-1} - \frac{1}{4}\Delta_{210}^{-1} \\
\big) &
\end{aligned}
$$

$$
\begin{aligned}
E = \zeta^2 \big(& \\
& + \frac{1}{24}\Delta_{14}^{-1} + \frac{1}{8}\Delta_{15}^{-1} - \frac{1}{16}\Delta_{34}^{-1} - \frac{3}{16}\Delta_{35}^{-1} - \frac{1}{48}\Delta_{49}^{-1} \\
& - \frac{1}{16}\Delta_{50}^{-1} - \frac{1}{24}\Delta_{64}^{-1} - \frac{1}{8}\Delta_{65}^{-1} + \frac{1}{8}\Delta_{111}^{-1} + \frac{3}{8}\Delta_{112}^{-1} \\
& + \frac{1}{8}\Delta_{121}^{-1} + \frac{1}{16}\Delta_{164}^{-1} + \frac{3}{16}\Delta_{165}^{-1} + \frac{1}{48}\Delta_{169}^{-1} + \frac{1}{16}\Delta_{170}^{-1} \\
& - \frac{1}{8}\Delta_{181}^{-1} - \frac{3}{8}\Delta_{182}^{-1} - \frac{1}{8}\Delta_{191}^{-1} + \frac{1}{4}\Delta_{206}^{-1} + \frac{1}{4}\Delta_{210}^{-1} \\
\big) &
\end{aligned}
$$